3-D PRINTED FOOD FEEDING SYSTEM

A PROJECT REPORT PHASE- I

submitted by

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in partial fulfilment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY



DEPARTMENT OF ELECTRONICS ENGINEERING

SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS)

KOTTAYAM

DECEMBER 2022

CERTIFICATE

Certified that this report entitled **3-D PRINTED FOOD FEEDING SYSTEM** is the report of Project Phase-I work done by **ASHISH THOMAS (MGP19BRABA014) ASTITAV SHARMA (MGP19BRBA015) THOMAS JACOB (MGP19BRBA044)** during the academic year **2022-2023** in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in **Robotics and Automation** of the Saintgits College of Engineering (Autonomous) under the APJ Abdul Kalam Technological University.

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ACKNOWLEDGEMENTS

We express our profound gratitude to all those who have supported us. Our sincere thanks to:

Er. Sita Radhakrishnan, under whose inspiring guidance and supervision this academic work was carried out. His insistent attitude and timely help led to the successful completion of Project Phase-I.

Er. Nishanth P R, for providing timely instructions and fruitful suggestions as the Project Co-ordinator.

Dr. Pradeep C, for providing valuable suggestions to improve the quality of our project.

Dr. Sreekala K S, Head, Department of Electronics Engineering for providing all the departmental facilities.

Dr. Sudha T, Dean, Department of Electronics Engineering for her kind support and motivation.

Dr. Josephkunju Paul C, Principal, Saintgits College of Engineering (Autonomous) for providing an excellent ambiance that led to the completion of this work on time.

Our colleagues, all the staff and students of the Department of Electronics Engineering for rendering help and support.

Our parents, friends and well-wishers for their constant support and timely help.

Special thanks to **Er. Harinarayanan Nampoothiri M.G.**, for introducing us to LaTeX document preparation system and providing us with his valuable support round the clock.

Above all we thank the Almighty God for His Grace and Blessings that made it possible for us to complete this Project Phase-I successfully.

ABSTRACT

KEYWORDS: Aging population ; Automated System ; RULA Analysis , Machine Learning; Assistive Robots

The world is now experiencing many aged societies. The definitions of an aged society are multiple and can be categorized into many subgroups, namely, aged, complete-aged, and super-aged societies. An aged society refers to a state where people equal to or more than 60 years constitute over 10% of the total population. Complete-aged and super-aged societies contain a volume of elderly people aged over 60 by more than 20% and 28% of the total population. Thailand is one among many countries facing aging society challenges. The country has been considered an aged society since 2005, and it may turn to a complete-aged and super-aged society by 2025 and 2030 respectively if the growth rate of aging population continues at the current pace. Considering the population of India and the rate at which the population is increasing it can be estimated that by 2030 it can also have an issue of having a large number of old people. A state of dependency of the elderly is one of the critical challenges in the health service system. This challenge is coupled with the high prevalence and incidence of chronic non-communicable diseases (NCDs) such as hypertension, diabetes, stroke, and coronary artery diseases. These diseases are likely to undermine the well-being of the elderly and make the elderly prone to a state of complete dependency, so-called, bedridden. A person who is bedridden or is unable to move portions of his/her body faces a lot of issues in day-to-day life, some issues can be resolved with little external help but some issues such as eating and going to the washroom will mostly require help from other human beings or helpers, this can be exhausting as well as frustrating for both the helper as well as the patient. To solve one such problem of feeding food we tried to make 'An automated food feeding system'. An automated food-feeding system is a type of machine that feeds food to a person when the command is given, this command can be given in many ways like a button, voice activation, etc. These systems can be used in a variety of settings, including homes, hospitals, shelter homes, etc. and can be designed to feed a wide

range of people having different medical conditions. There are several benefits to using an automated food-feeding system. For one, it can help to ensure that person receive a consistent and balanced diet, as the system can deliver the correct portion size each time. Additionally, an automated system can be more convenient for hospitals or caretakers, as it can save the time and effort that would be required to manually feed the animals. In some cases, automated systems may also be able to reduce food waste, as the amount of food that has to be fed can be more accurately controlled. There are several different types of automated food-feeding systems available, including those that use gravity-fed hoppers, electronic timers, and sensors to detect presence and position of the person. Some systems may also be connected to a computer or smartphone, allowing users to remotely control and monitor the feeding process. In our project, we tried to make such food feeding system that can be used by all age groups, easily available as well as easy to set up and use. To make such systems different techniques such as RULA analysis, sensor-based machine learning techniques we studied. Simulation was done by using these methods and techniques to finalize the best suitable way to make such system. Research papers were studied to get basic ideas of how these system works. Overall, food-feeding robots can be a convenient and efficient way to provide meals and other items to bedridden individuals, helping to reduce the burden on caregivers and improve the quality of life for the person receiving care. We decided to work on this project because we believe anything that brings people's dignity back is a great step for the betterment of society as well as the whole world. It's worth noting that these types of robots are still in the early stages of development and may not yet be widely available. Additionally, they may require the assistance of a caregiver to set up and operate.

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ABBREVIATIONS

TRL	Technology Readiness levels
lbs	Pound-Mass
LRF	Laser Range Finder
PC	Personal Computer
ICRAFT	Eye-Controlled Robotic Feeding Arm Technology.
GUI	Graphical User Interface
CAD	Computer-Aided Design
DOF	Degrees of Freedom

CHAPTER 1

INTRODUCTION

Assistive robots with which users can directly interact have attracted worldwide attention. Assistive robots can assist people with disabilities and older persons in the activities of daily living. For improving quality of life and provisioning for demographic shifts, assistive robots could be one of solutions. The customizing users and their culture is one of crucial issues. In addition, the cost-effectiveness is important issue as well. Several kind of assistive robots have been proposed: self-feeding robots, workstation-based robots, powered wheelchair-based robots, mobile robots, exoskeletons, entertainments and monitoring robots and so on

In order to support care-giving tasks for the people with disabilities and older persons, caregivers should physically interact with the people. For example, caregivers support the persons to have a meal, change their clothes, reposition their posture, transfer their locations, and take a bath. Other tasks in the care of the disabled and the elderly could include activities of daily living. Among these activities, having a meal is one of essential activities for their living and caregivers should interact with the people frequently, i.e., kind of food, and the feeding interval. Having a meal could be replaced with existing robotic technologies. Assistive robots are solutions to deal with supporting the activities of daily livings

1.1 Review of Meal Assistance Robots

Primary objective of a meal assistance robot is to assist or help in self-feeding process of a particular individual who do not have enough capacity to feed himself/herself using his/her upper-limb. Even though there are no standardized components of a meal assistance robot, it should consist of a robot arm or feeding mechanism, control hardware to drive the robot or mechanism, a command algorithm and sensors to capture the human motion intension

The Most Common meal assistance Robots are:

1. My Spoon
2. Winsford Feeder
3. Meal Support System
4. ICRAFT

1.1.1 My Spoon

The mealtime robot My Spoon, developed by the Japanese company SECOM , focuses on people who are unable to eat without the help of others (e.g., due to limited arm/ hand function). It is a robot that provides meal assistance by providing a special plate or bowl that keeps the food and a robot arm with a fork and spoon attached that picks up the food and delivers it to the user. My Spoon allows the user to eat with only minimal help from a caregiver, at his or her own speed, and according to his or her own preferences. It creates independence and reduces the amount of care. My Spoon is a well-established commercially available robot at a TRL of 9. My Spoon supports the user physically with the eating activity without any social interactions.

1.1.2 Winsford Feeder

Winsford feeder help people who have lost the ability to feed themselves. All that's required to operate the Winsford Feeder is a slight move of the head. The Winsford Feeder can also be controlled with other parts of the body such as hands, shoulders, hips or knees using the rocker switch. The Winsford Feeder includes a carrying case, battery, charger, dinner plate, bowl, spoon, pusher, chin switch rod, stationary glass holder and turntable. It also includes a shelf and drip pan. Feeder measures 12" x 18" (51 x 46cm) and weighs 18 lbs. (8.2kg). An extra rechargeable battery is available separately. Using the charger during operation keeps the battery charged. The feeder also works without the battery. The height of the feeder is adjusted with the large knob. The Winsford Feeder is designed for persons who are able to sit in an upright position. The feeder requires a minimum height of 10" (25cm) from the tabletop.

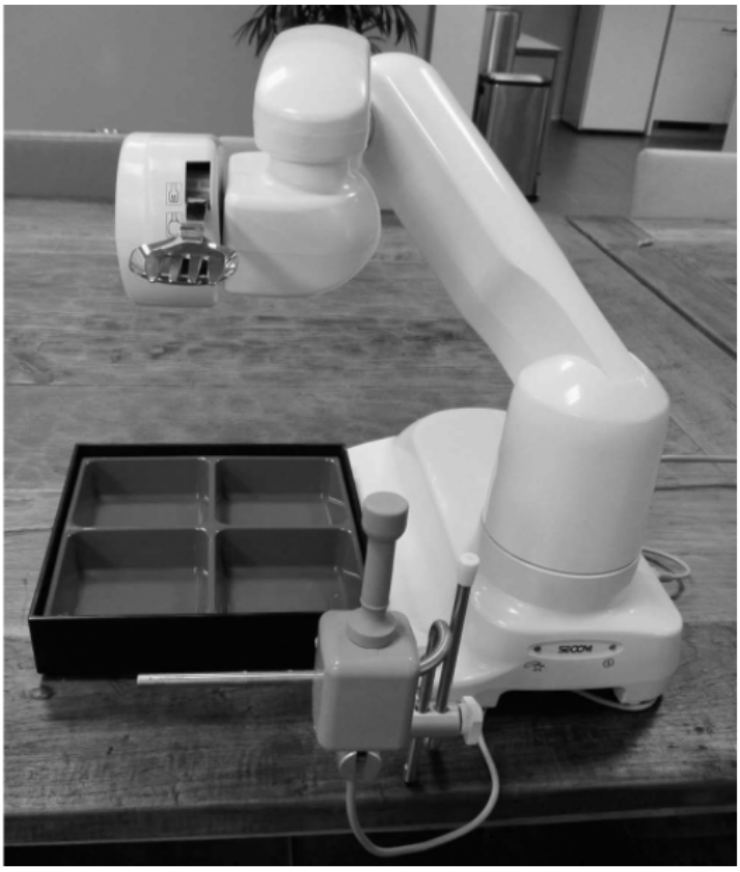


Figure 1.1: My Spoon Robot



Figure 1.2: Winsford Feeder

1.1.3 Meal Support System

The robot system is equipped with a laser range finder (LRF) and a manipulator holding a spoon. The LRF measures the 3D coordinates of surface points belonging to food on a plate. Then the robot determines the position of food surface to scoop, and the manipulator moves according to the calculated trajectory. The system has an advantage that preparation of food cutting in bite-size is not required. The proposed scooping control was implemented and verified in experiment with two kinds of non-rigid solid foods. It was shown that the robot can scoop foods for the most part with high success rate. here are an LRF fixed above food on a plate, a manipulator, a spoon attached at the tip of manipulator and food on a plate. In order to reduce the noise with measurement, a plate and a table that lack luster are used. A plate is not fixed on a table, which implies that the plate might move slightly by each scoop- ing motion. The LRF and manipulator are connected to a PC. The LRF sends 3D point data to PC. The command sent from PC to manipulator provides trajectories of the tip of the manipulator. Two springs are attached between the spoon and the hand tip, as depicted in Fig. 2. This me- chanical compliance can soften the force between spoon and plate.

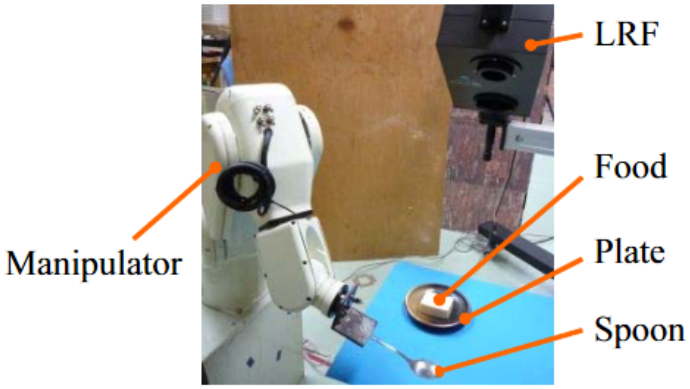


Figure 1.3: Meal Support System

1.1.4 ICRAFT

ICRAFT consits of an interface with the eye-tracking technology occurs with a computer monitor with the webcam mounted on top of it. The camera has an IR light that shines in the users eye and which allows the eye- tracking software to monitor the

movement of the users eye. This program will seize control of the computer, acting as a mouse would for a normal computer and allow the user to select from the three bowl positions and initiate and conclude the feeding process. The feeding process works in the following manner, with the user first defining the feeding position, or the location of the users mouth where the food is to be brought to. The user then selects between the three food locations corresponding to different foods with the eye-tracking GUI appearing on the monitor behind the arm. The robot arm then acts out a scooping and feeding motion through preprogrammed sequences. This ensures the maximum amount of smoothness and a comfortable feeding speed to be achieved. The automatic program will bring the food to the defined feeding location so the user can move their head forward slightly and eat the food off the spoon. In between food selections the robot arm will return to a rest position, which is also the starting position the robot arm will be in upon first powering the unit up. The user also has the opportunity to select drinking while they finish their bowl selection.

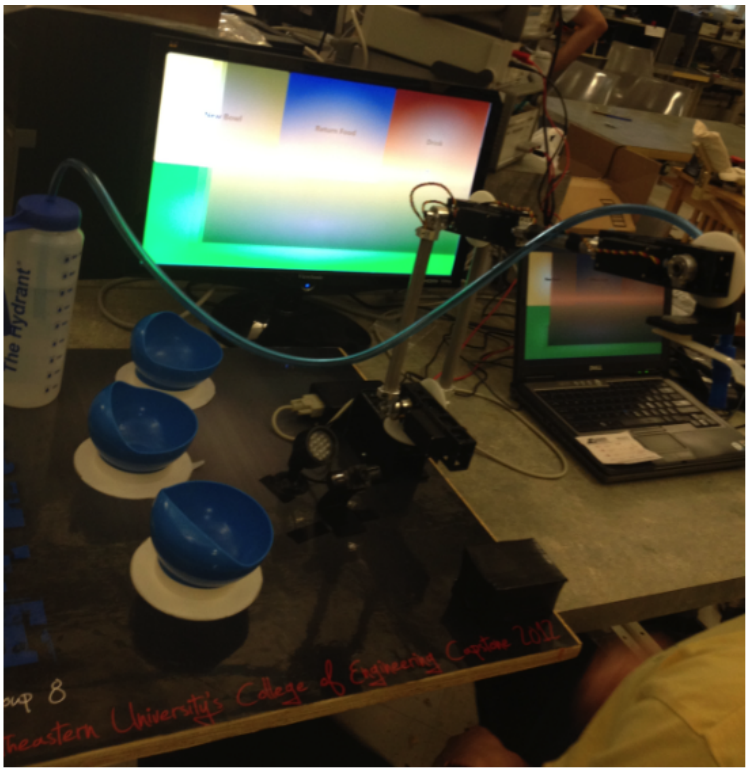


Figure 1.4: ICRAFT Robot Setup

CHAPTER 2

LITERATURE REVIEW

2.1 Daehyung Park a, Yuuna Hoshi a, Harshal P. Mahajan b, Ho Keun Kim a,Zackory Erickson a, Wendy A. Rogers c, Charles C. Kemp: "Active robot-assisted feeding with a general-purpose mobile manipulator: Design, evaluation, and lessons learned." (2019)

This paper shows that General purpose mobile manipulators were used as feeding arms . Different arms having different DOF was used to get to know how many DOF will be basic requirement for the robot to work efficiently and to get to the finishing point i.e mouth if minimum time and minimum movement. Machine learning techniques along with the camera were used to find the mouth of the person.Completion time of the task were also noted for different types of foods such watermelon,cereals etc.

2.2 Schultz, J. R., Slifkin, A. B., Yu, H., Schearer, E. M. . Proof-of-Concept: A Hands-Free Interface for Robot-Assisted Self-Feeding. (2022)

This paper involves building a prototype system that demonstrates the feasibility and functionality of the proposed interface. The prototype could be tested with a small group of users to gather feedback and assess the performance of the system. This could involve conducting user studies to evaluate the ease of use and effectiveness of the interface by developing software to control the movement of the robot and interpret commands from the user. as well as identifying any potential issues or challenges that need to be addressed.

2.3 Daehyung Park; Hokeun Kim; Yuuna Hoshi; Zackory Erickson; Ariel Kapusta;Charles C. Kemp: "A multimodal execution monitor with anomaly classification for robot-assisted feeding" (2017)

This paper deals highlights a multimodal execution monitor with a robot-assisted feeding system that performs scooping and delivers a spoon of yogurt to the mouth of participants was used and studied. Robot was presented with different anomalies to see how the system reacts when person moves from its previous position,when the type of food changes etc. Basic pattern of anomalies were studies to further train the robot, overall effectiveness, or accuracy, of the system was also studied and improved .

2.4 Ethan K. Gordon; Xiang Meng; Tapomayukh Bhattacharjee; Matt Barnes; Siddhartha S. Srinivasa: "Adaptive Robot-Assisted Feeding: An Online Learning Framework for Acquiring Previously Unseen Food Items" (2020)

The robot was made to pick different food items with same end effector the based on the results different end effectors were used for different types of food .Foods such as banana, grapes and apple slices all have different structure and softness, the robot was taught how to apply pressure and angle of contact to pick these food items. An adaptive robot that can be used for variety of food is what they were trying to make.

2.5 Patrick Deegan,Roderic Grupen,Allen Hanson,Emily Horrell, Shichao Ou,Edward Riseman,Shiraj Sen, Bryan Thibodeau,Adam Williams Dan Xie: "Mobile manipulators for assisted living in residential settings" (2007)

This paper mostly taught how machine learning is used along with different sensors to reach the final position and achieve the goal it is intended to do . Different algorithms were used to do different tasks by same robot.

2.6 Veronique Maheu; Philippe S. Archambault; Julie Frappier; François Routhier: "Evaluation of the JACO robotic arm: Clinico-economic study for powered wheelchair users with upper-extremity disabilities" (2011)

Evaluation of JACO robotic arm was done in this paper. JACO is a 7 DOF Robotic arm, . The 7 degree-of-freedom refer to 6 movement in three-dimensional space (up, down, left, right, back and forth - 3 DOF), 6 movement of JACO's wrist (abduction, adduction, flexion, extension, pronation, supination - 3 DOF) and opening and closing of the 3 fingers (1 DOF). These movements of the arm were studied as a reference to be used in making the robotic manipulator that can feed a person.

CHAPTER 3

OBJECTIVES

The ultimate aim of this proposed system is to assist the persons with movement disorder who need the help of someone to feed themselves. The main goals of this project are

- Make a device that can help the needy.
- Easy to use.
- Make it as cheap as possible so that everyone can use it.
- Easily available.
- Recognise anomalies and provide efficient assistance.

The main objectives of this project are:

Range of Use

The main objectives that we had when we started this project was only to make a human feeding robot that can help them to have their food without the need of any caregiver but as we moved along the way to do that, many other objectives also came that we should do in order to efficiently achieve the final goal . The first objective was to made the system in a way that it can be used by people having different medical problems. Some people are unable to speak so voice activated system could not be used. Some people have shivering hands so its difficult for them to use touch screen or sensitive buttons.

Easily available and cost efficient Material

The second objective was to made the robotic system durable as well as cost efficient ,so the basic materials such as steel ,iron etc. could not be used as they are bit costly.

Ergonomic System

Third objective is needed to be achieved is to make the system ergonomic as much as possible because the system will mostly be used by the people having medical conditions, sharp corners and edges should be avoided.

Reduce Workload and Stress

Fourth objective that must be covered in this project is to make the system as independent as possible so that caregiver person can just setup or turn on the system and go to some other works. He/She should not have to stand there when the system is working .

Feasibility

Fifth objective was to keep the working of the system as simple as possible , there should not be some advance commands that should be given to the system to make it work. A normal push button should be able to start that system

CHAPTER 4

PROJECT PLAN AND TASK ALLOCATION

The project is divided into two phases;

4.1 Phase 1

Phase 1 deals with thorough literature study, designing the 3D model in Solidworks and simulation using a miniature model for better understanding the working and helps in the further development of large model.

As per our current scope we have mapped out our phase 1 strategy as shown below. The table also shows our progress so far.

Duration of week / Name of Task	Week 1	Week 3	Week 5	Week 7	Week 9	Week 10	Week 11	Week 12
Literature Survey								
Study of Components								
Structure Design								
Design & Simulation in Software								
Report Writing								

Table 4.1: Project Plan for Phase I

The following are the list of resources we require:

1. Solidworks
2. Overleaf Document Preparation Tool

The CAD model is designed in Solidworks software and the final reporting is done using overleaf. The various the tasks associated with this project is allocated as follows:

Ashish Thomas	**Astitav Sharma**	**Thomas Jacob**
Study of Components	Study of Components	Study of Components
Literature Survey	Literature Survey	Literature Survey
Structure Design		Structure Design
Design & Simulation		Design & Simulation
PowerPoint Presentation	PowerPoint Presentation	
Report Writing	Report Writing	Report Writing
		Project Diary Entry

Table 4.2: Task Allocation

4.2 Phase 2

Phase 2 deals with the design and fabrication of the model and testing of the prototype.

CHAPTER 5

DESIGN AND METHODOLOGY

5.1 Design

The designing of the arm is done in the CAD software, Solidworks. The individual parts are modeled separately using component dimensions and finally all parts are assembled.

1. Individual parts

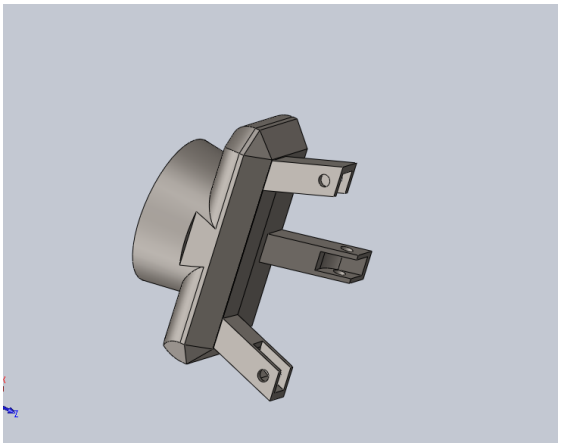


Figure 5.1: Gripper

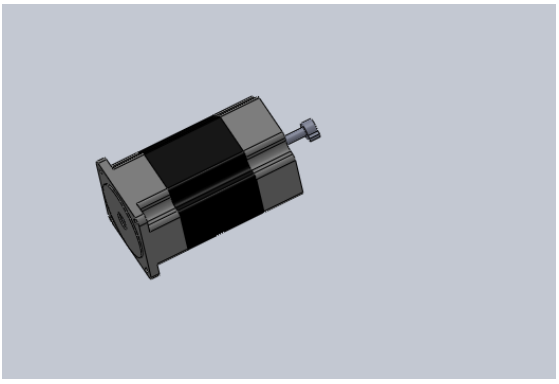


Figure 5.2: Motor

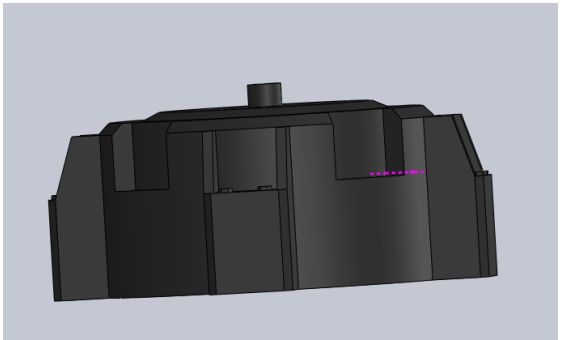


Figure 5.3: Base-Front View

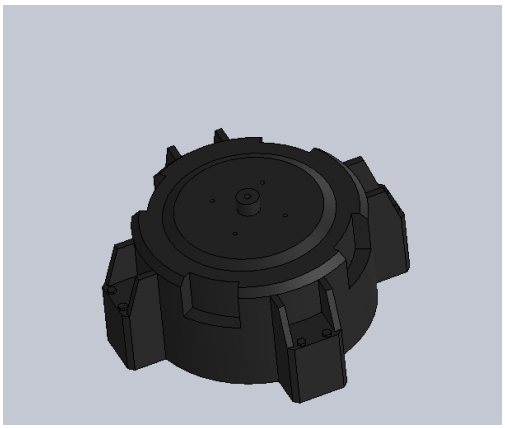


Figure 5.4: Base-Top View

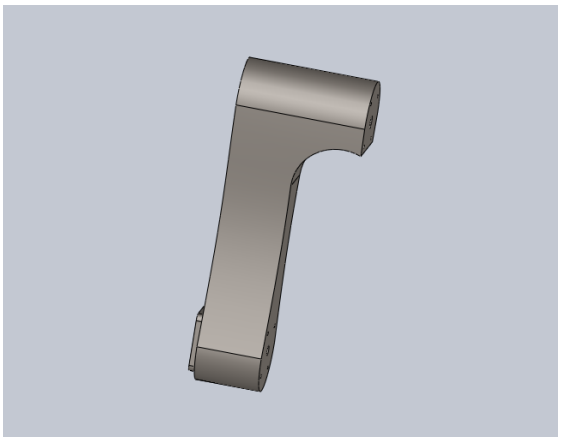


Figure 5.5: Second Arm

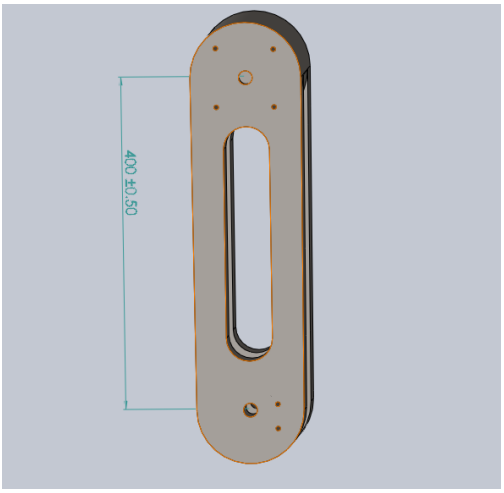


Figure 5.6: Third Arm

2. Final Design

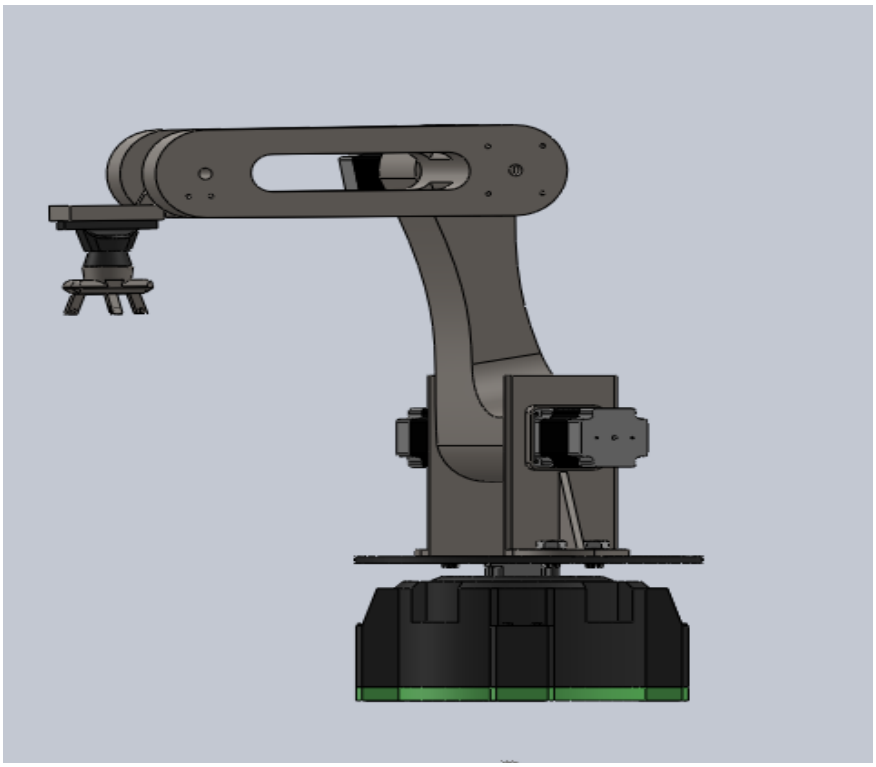


Figure 5.7: Final Design

5.2 Methodology

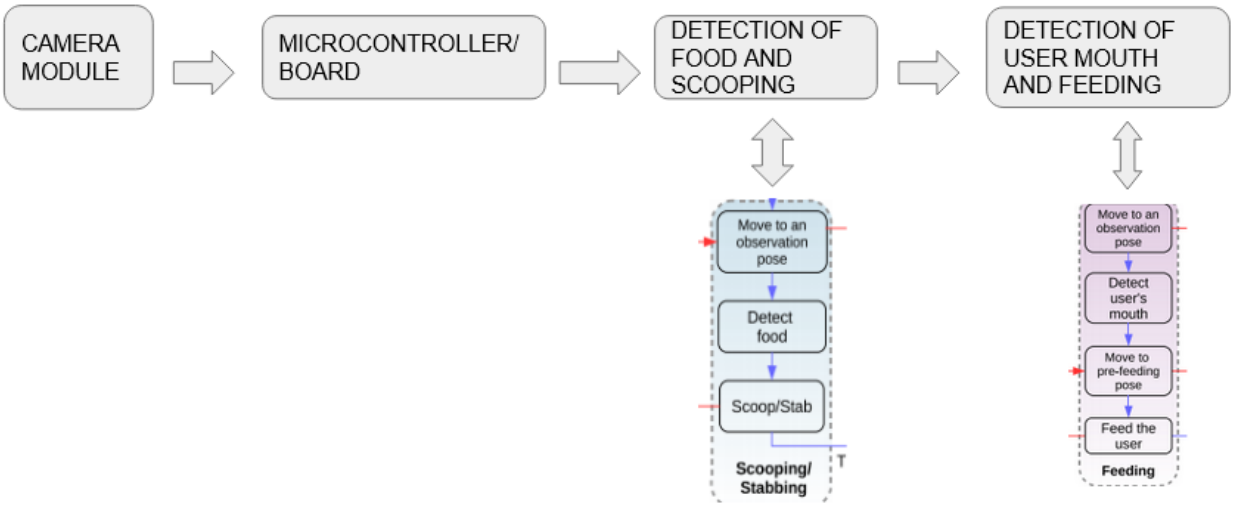


Figure 5.8: Simpilified Block Diagram

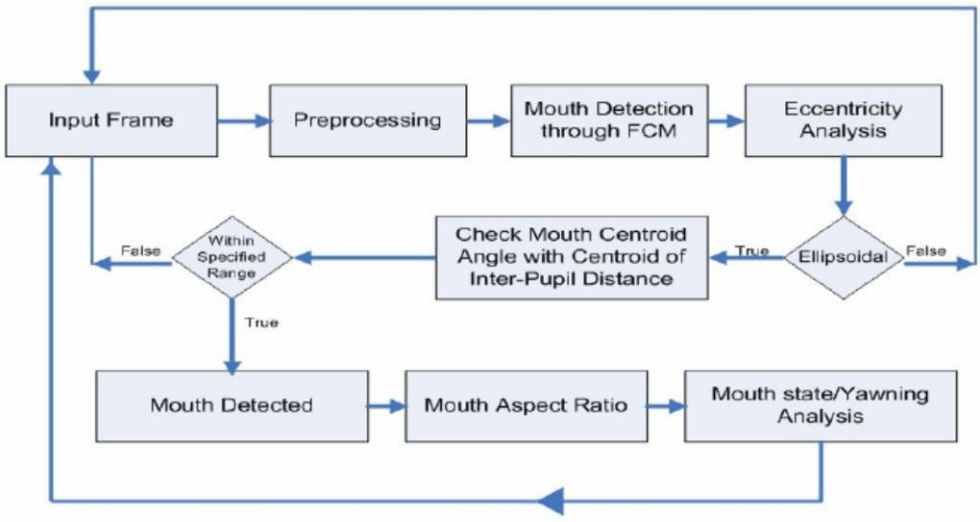


Figure 5.9: Mouth Detection Flowchart

CHAPTER 6

PRELIMINARY RESULTS

Simulation was done in Solidworks Software

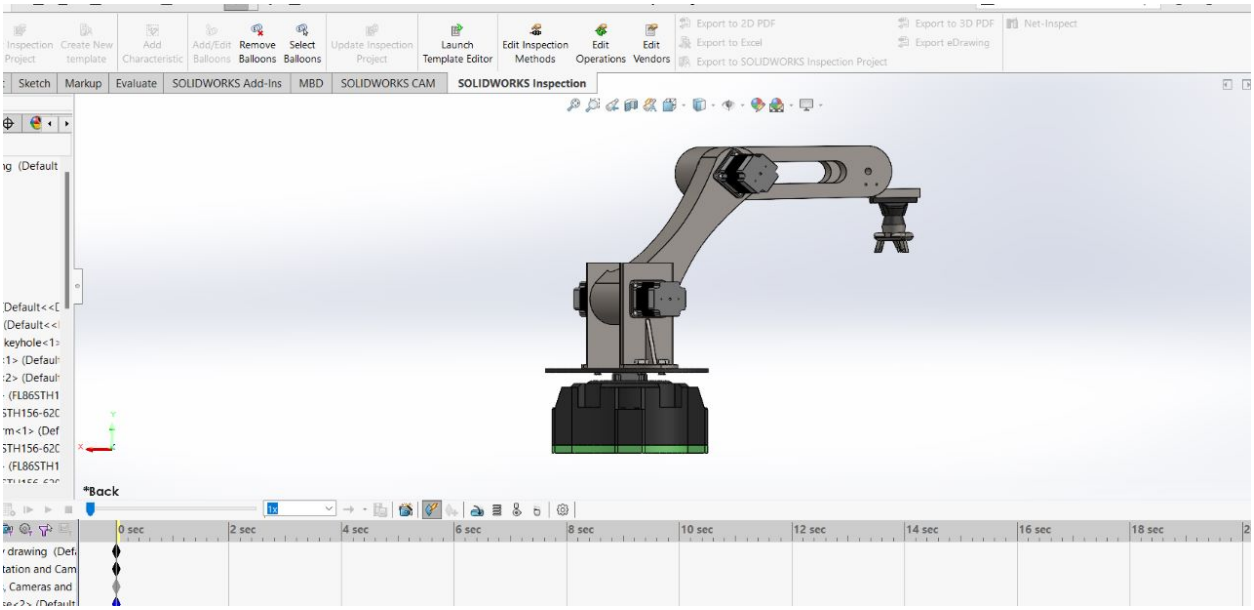


Figure 6.1: Preliminary Result

CHAPTER 7

CONCLUSION

The system we have designed is focused on assistance rather than rehabilitation. Automated assistance can enable people with diverse disabilities to perform daily activities by themselves. However, robotic assistance could also potentially discourage people from using their physical abilities in activities that they could perform with less assistance. There is also the potential for robotic assistance to affect peoples' rehabilitation status or worsen their health through reduced physical activity. These are risks that merit attention as robotic assistance becomes more common. In addition to assistance, general-purpose robots could potentially help during rehabilitation or modify their assistive actions to encourage exercise by users.

During Project Phase II we are planning to construct the prototype of a proposed system, we will progress the usability test of a proposed system, thus improve the overall system and for the comfortable use in accordance with usability tests. Hopefully this system will have the potential to reduce family caregivers' prolonged stress, physical demands, and decline in quality of life while reducing the healthcare costs for households with people with disabilities.

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